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The transference of *Mehrarhynchus* species (Acanthocephala) to the genus *Cleaveius* is questionable? - a review

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Abstract : The transference of the species belonging to the genus *Mehrarhynchus* to the genus *Cleveius* by Golvan (1959)[1] and Bhattacharya (2007)[2] is unwarranted. The authors have reasons to consider the genus *Mehrarhynchus* valid & supported by Yamaguti (1963)[3], who considered genera *Cleaveius & Mehrarhynchus* separate with suitable scientific reasons. The authors have reasons to place ***M.manipurensis*- a species under enquiry**.

Key words : ***Cleaveius, Mehrarhynchus.***

INTRODUCTION

The genus *Cleaveius* (Micracanthorhynchinidae)* was erected by Subrahmanian (1927)[4] to include *C.circumspinifer* recovered from an undetermined fresh water fish in Burma, whereas the genus *Mehrarhynchus* was erected by Datta (1940)[5] with type species *M.prashadi* harbouring *Pangasius pangasius* at Kolkata. The genus *Mehrarhynchus* was considered by Golvan (1959)[1] & Amin (1985)[6] to be synonym of *Cleaveius* to which Jain (2003)[7] did not agree rather considered *Mehrarhynchus* an independent genus.

Yamaguti (1963)[3] held both the above mentioned genera valid. No other species was added to the genus *Cleaveius* (except a few) since its erection but in the genus *Mehrarhynchus* a number of species were described since its inception. The authors of the *Mehrarhynchus* species considered the genus *Mehrarhynchus* valid. The Indian species of the genus *Mehrarhynchus* are:-

1. *M.prashadi* (Datta, 1940)[5] Golvan (1969)[8].
2. **M.secundus* Tripathi (1959)[9] harbouring (intestine of *Pama pama, Osteogeneiosus militaris*, locality being Malta river, W.B. and *Plotosus canis* from Chilka lake (Orissa). It was also dealt by Golvan (1969)[8].
3. *M.mysti* Sahay & Sinha (1971)[10] harbouring *Mystus seenghala*, intestine.
4. *C.leiognathi Jain & Gupta* (1979)[11].
5. *C.portblairensis* Jain & Gupta (1979)[11]
6. *C.fotedari* Gupta and Naqvi (1982)[20].
7. *M.inglishi* Gupta and Fatma (1987)[12]. Also dealt by Kumar (1992)[13]. Harbours Mugil tade, in the intestine locality being Quilon, Kerala.
8. *Mehrarhynchus singhai* (Gupta & Fatma, 1987)[12] Kumar, 1992[13] in *Osteogeneious militaris* intestine, locality being Quilon, Kerala.
9. *C.puriensis* Gupta & sinha (1992).[14]
10. *C.clupeia* Gupta & sinha (1992).[14]
11. *M.manipurensis* Zenith, Gambhir, Lakshmipyari & Gyaneshwari (2012)[15].

A comparison of the families like *Micracanthorhynchinidae* Yamaguti(1963)[3] and *Rhadinorhynchidae* Travassos (1923)[16] (Table I) reveals that in both families – Trunk is spinose but in

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Rhadinorhynchidae trunk spines are either extensive or limited (in ant part only), proboscis hooks are very numerous, cement glands are 2-8 usually elongated where as in *Micracanthorhynchinidae* trunk spines are though limited on anterior portion & proboscis hooks are usually compact, exceptionally tubular; proboscis hook are nearly uniform in shape though different in size & ventral proboscis hooks are as large as dorsal hooks (vide page 36-37 of the key to the families of *Echinorhynchidea* Yamaguti (1963)[3].

A comparison of the genera (Table II) depicts that (1) Arrangement of Trunk spines are in two group (anterior & posterior in *Cleaveius* is not found in *Mehrarhynchus* where trunk spines have been found to be situated in anterior region with regular rows of small triangular spines **Besides the ant group is absent in male of *Cleaveius***. (2) Proboscis in *Cleaveius* is short cylindrical with 18 longitudinal rows of few (four) hooks each whereas in *Mehrarhynchus* proboscis is long with numerous fine recurved hooks (12-14 longitudinal rows of 20-22 hooks each). (3) Leminsci in *Cleaveius* are long winding whereas in *Mehrarhynchus* though the leminsci are long, filamentous may be bifurcated at the end (in *M.secundus*). (4) ganglion is situated in *Cleaveius* lateral to the base of proboscis receptacle where as it is median slightly above the base of receptacle in *Mehrarynchus*.

It seems that Golvan (1959)[1] did not accept the genus *Mehrarhynchus* & thus merged it with *Cleaveius* Subrahamanian (1927)[4] without giving a reasonable ground & knowing fully well that Yamaguti (1963)[3] had recognized both the genera valid.

It is surprising that in 2007 Bhattacharya supported Golvan (1959)[1] in the merger of the genus *Mehrarhynchus* Datta (1940)[5] with *Cleaveius* without giving a plausible reason and also provided a key to the species of *Cleaveius* transferring all the species reported under the genus *Mehrarhynchus.*

It seems that Zenith *et.al.* (2012)[15] did not accept the merger of *Mehrarhynchus* with *Cleaveius* and that is why she along with her colleagues described *Mehrarhynchus manipurensis* n.sp. in 2012.

The present authors vehemently protest the merger of the genus *Mehrarhynchus* with *Cleaveius* (vide comparative chart of the genera- Table III) and on points mentioned earlier (point I to 4).

Besides above, *Mehrarhynchus* Datta (1940)[5] belongs to the family *Rhadinorhynahidae* (*Echinorhynchidea*) where as *Cleaveius* belong to a different family viz *Micracanthorhynchinidae*, *Echinorhynchidea*). These two families are quite separate. In *Micracanthorhynchinidae*, anterior proboscis hooks have bifurcate root, trunk spines are more extensive ventrally than dorsally & usually not divided in two groups except in *Cleaveius* where trunk spines are divided into two groups at least in female and ganglion is situated a little anterior to base of proboscis receptacle contrarily in *Rhadinorhynchidae* (Luhe, 1912[12] & Yamaguti, 1985)[18] trunk is long slender or swollen in anterior or middle third, spined extensively or on anterior part only, proboscis receptacle is cylindrical with gangulion at or near base or equatorial, testes are contiguous or not usually in middle third of trunk, cement gland 2-8 globular or pyriform.

CONCLUSION

On the basis of the arguments stated above transference of species of *Mehrarhynchus* to the genus *Cleaveius* is unwarranted & the authors restore all the species of *Mehrarhynchus* and all species except a few are held valid under the genus *Mehrarhynchus.*

As regards *M.manipurensis* Zenith *et.al.* (2012)[15], it is suggested that the authors should re-examine the "Single layered proboscis receptacle" as this point is against the generic character, till then the **species is kept under enquiry**. Zenith *et.al.* (2012)[15] has described the presence of saefftingen pouch in *M.manipurensis* but in diagram it has not been shown. Zenith *et.al.* (2012)[15] though mentions in case of *M.manipurensis* proboscis to have 8-10 longitudinal rows of hooks but forgot to count their number in each row. The measurement of right & Left ovarian ball is not correct. It should be only ovarian balls. The authors of the said species mention that there is no bursal cap in male & uterine bell in female, uterus is simple. The concept of absence of uterine bell is rather doubtful,

*Bhattacharya (2007)[18] has suggested generic modification of *Cleaveius* & advocated inclusion of "egg with polar prolongations of the middle shell" in the genus *Cleaveius*

it can only happen in post coitus period or in extreme early stage of life cycle. Bursal cap is a must for retraction and extension of bursa from cap during copulation. The authors suggest re-examination of the species in question.

Table 1: Comparative chart of *Micracanthorhynchinidae* Yamaguti (1963)[3] and *Rhadinorhynchidae* Travassos (1923)[16] : *Rhadinorhynchinae* Luhe (1912)[17]

	Echinorhynchidea Southwell *et.* Macfie (1925)[19] *Micracanthorhynchinidae* Yamaguti (1963)[3]	*Echinorhynchidea* Southwell *et.* Macfie (1925)[19] *Rhadinorynchidae* Travassos (1923)[16]
Proboscis	Claviform to cylindrical with hooks nearly uniform in shape, larger anterior hooks with bifurcate root.	Long or short, cylindrical
Probscis receptacle	-	Usually cylindrical
Ganglion	- At the base or middle of proboscis receptacle	At or near base or equatorial
Leminsci	-	Variable in length, longer or shorter than proboscis receptacle
Trunk/ Trunk spines	Generally more extensive ventrally than dorsally usually not divided into two groups	Long and slender or swollen in anterior or middle third, spined extensively or on anterior part only, occasionally at post. extremity too, the spines may be lost secondarily and present in young individuals alone
Hypodermic nuclei	-	Large & few or small & numerous
Cement gland	4-8 compact pyriform to claviform exceptionally tubular	2-8, globular pyriform, claviform or cylindrical or very long, slender or elongate.
Testes	-	Contiguous or not usually in middle third of trunk, occasionally anterior
Proboscis hook	Not numerous	Are very numerous
Eggs	-	Elongate or oval
Parasites of	Fishes	-Fishes

Table 2:

Family	*Micracanthorhynchinidae*	*Rhadinorhynchidae* Travassos (1923)[16] *Rhadinorhynchinae* Luhe (1912)[17]
Genus	*Cleaveius*	*Mehrarhynchus*
Author	Subrahmanian (1927)[4]	Datta (1940)[5]
Body	Medium sized cylindrical	Small
Trunk spines	Divided into two groups (i) anterior group- irregularly distributed (ii) posterior group in numerous transverse rows. In male anterior group is lacking but in females the anterior two thirds of body are armed with approximately 41 circular rows of about 14 spines each	Armed in its anterior region with regular rows of small triangular spines in both sexes.
Proboscis	Very short, cylindrical with about 18 longitudinal rows of few (4) hooks each	Long, cylindrical with numerous fine recurved hooks (14-14 longitudinal rows of 20-22 hooks each.
Neck	-	Short

Contd..

Table 2 Contd..

Proboscis receptacle or sheath	Long, double walled	Proboscis sheath with double layered walls
Ganglion	Situated laterally a little anterior to base of proboscis receptacle.	Situated in wall of proboscis sheath a little anterior to its posterior end.
Leminsci	Very long, winding	Long, filamentous, may be bifurcated at the end (in *M.secundus)*.
Cement gland	4 compact pyriform to tubular	4, clavate
Testes	-	Directly tandem
Uterine bell	-	Funnel shaped, uterus narrow
Vagina	-	With dumb-bell shaped sphincter
Vulva	-	Posteroventral
Eggs	-	Not observed
Parasitic in	Fishes	Fishes

Key to the species of the genus *Mehrarhynchus* Datta (1940)[5] as proposed by Gupta & Fatma (1987)[12]
***Indian. Jour. Helminthol.* 39(1):** 1 – 26**.**

Lemnisci equal	- 1
Lemnisci unequal	- 3
1. Lemnisci bifurcated at free end	- *M.prashadi* Datta (1940)[5]
Lemnisci not bifurcated at free end	- 2
2. Proboscis with 18-20 hooks per row	- *M.singhai* Gupta & Fatma (1987)[12]
Proboscis with 15-17 hooks per row	- *M.mysti* Sahay & Sinha (1971)[10]
3. Testes post equatorial, lemnisci smaller than receptacle	- *M.secundus* Tripathi (1956)[9]
Testes equatorial, lemnisci larger than receptacle	- *M.inglisi** Gupta & Fatma (1987)[12]

Key to the species of the genus *Cleaveius* Subrahmanian (1927)[4] as proposed by Bhattacharya (2007)[2] in a hand book on Acanthocephala 2007.

1. Leminsci bifurcated	- *C.prashadi* Datta (1940)[5]
Leminsci not bifurcated	- 2
2. Proboscis hooks in 12-14 x 16-18	- *C. inglishi* Gupta & Fatma (1987)[12]
Proboscis hooks more	- 3
3. Proboscis hooks 14 – 18x 15 – 17	- *C.mysti* Sahay & Sinha (1971)[10]
Proboscis hooks more	- 4
4. Proboscis hooks 18-20 x 16-18	- *C.secundus* Tripathi (1956)[9]
Proboscis hooks 16 – 18 x 18 – 20 with polar prolongations of middle shell membrane	- *C.singhai* Gupta & Fatma (1987)[12]

*Gupta & Fatma (1987)[12] compared their collection of acanthocephalan (*Mehrarhynchus)* only with *M.secundus* Tripathi (1956)[9] & proposed n.sp. (*M.inglisi*) on differences (unequal lemnisci not bifurcated at free end, T_2 larger than T_1, equatorial testis). They should have compared their specimens with *M.prashadi & M.mysti* also before coming to a conclusion. T_2 larger than T_1 is also met in *M.prashadi, M.singhai*. Equatorial testes is also seen in *M.prashadi.* Similarly they compared *Myhrarhnchus singhai* with M (*secundus & mysti*) only. Why they forgot to compare with *M.prashadi* is rather not understandable.

Table3:

	M.prashadi (Datta, 1940) Golvan, 1969	*M.secundus* Tripathi, 1956	*M.mysti* Sahay & Sinha, 1971	*M.singhai* (Gupta & Fatma, 1987); Kumar, 1992	*M.inglisi* (Gupta & Fatma, 1987); Kumar, 1992	*M.manipurensis* Zenith *et.al.*,2012
Size of body Male Female	4.62 – 6.71x0.68-0.79 7.96x1.53	2.07-2.49x0.37-0.49 1.84-2.69x0.09-0.4	Absent 1.77-5.31x0.65-0.69	5.54-5.85x1.0-1.20 5.40-6.55x0.90-1.0	4.85-5.30x0.9-1.2 5.5-5.86x0.85-1.0	3.28-3.31x0.65-0.67 3.37-3.39x0.67-0.69
Proboscis size	0.95 x .33	0.55-0.79x0.244-0.304	0.80 x 0.93	1.0 – 1.05 x 0.3	0.80-0.82x0.32	0.48-0.49x0.19-0.20 ♀0.51-0.53x0.16-0.18
Longitudinal row of proboscis hooks	12-14 of 20-22 hooks in each	18 - 20	14 - 18	16 - 18	12 – 14	8 – 10 long row
Hooks per row	20 – 22	16 – 18 Roots longer than barb	15 – 17	18 – 20	16 – 18	14 – 16 rows transversely arranged. Ant. half of trunk has 25 -30 long row of spines ♂-29,♀-27)
Limnisci	Equal bifurcated at free end	Unequal not bifurcated at end	Equal not bifurcated at end	Equal, not bifurcated at end	Unequal, not bifurcated at end	Equal 1.41-1.43x0.07-0.09 ♂-0.91-0.93x0.08-0.10
Trunk spines		Ant.with 9-10 rows	-	-	8-10 circles	
Extension of leminsci	Longer than Pr.receptacle	Smaller than receptacle	Longer than receptacle	Longer than receptacle	Longer than receptacle	
Proboscis sheath	-	0.785x0.029			L/1-1.90-2.65; L/2-2.10 – 2.5	
Testes	Tandem, equatorial separate	Tandem post equatorial overlapping	Males absent	Tandem, post equatorial, overlapping	Tandem equatorial overlapping	Nature not mentioned
T_1	0.22x0.37	0.116 x 0.145 x 0.145 -0.188		0.40-0.50x.0.35	0.40-0.45x0.35	0.18-0.20x0.17-0.19
T_2	0.33x0.37	Equal T_1		0.45-0.55x0.30	0.50-0.55x0.36	0.19-0.21x0.17-0.20
Cement gland	4	4 pyriform		4 pyriform	4 pyriform	
Sacfftingen pouch	0.81 long	0.10 dia				0.25 x 0.33
Bursa						♂0.58-0.61x0.07-0.09 (bursal cap absent) ♀0.72-0.75x0.23-0.25
Uterus						
Egg	-	Not described	-	0.076-0.078x0.02-0.025 with polar prolongation of middle shell membrane	0.076-0.078 x 0.02-0.025	tubular & simple (uterine bell absent)
Host	*Pangasius pangasius*	*Plotsus canis, Pama pama*	*Mystus seeenghala*	*Osteogeniosus militaris*	*Mugil tade*	*Glyptothorax granulus*
Locality	Kolkata	Chilka Lake & Malta river Odisha	Ranchi, Bihar	Quilon, Kerala	*Quilon Kerala*	Maklang, Imphal West district.

Key to the Indian species of the genus *Mehrarhynchus* (Proposed)

1. Lemnisci equal - 2.
   Lemnisci unequal - 3
2. Lemnisci bifurcated at free end - *M.prashadi* Datta (1940)[5]
   Lemnisci not bifurcated at the free end - 4.
3. Testes post equatorial, lemnisci smaller than proboscis receptacle - *M.secundus* Tripathi (1956)[9]
   Testes equatorial, lemnisci larger than proboscis receptacle - *M.inglishi* Gupta & Fatma (1987)[9]
4. Probosis with 18-20 hooks per row, eggs filamented - *M.Singhai* Gupta & Fatma (1987)[12]
   Probosis with 15-17 hooks - *M.mysti* Sahay & Sinha (1971)[10]
   Probosis with 14-16 long row with (?) - *M.manipurensis** Zenith *et.al.* (2012)[15]

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*Species under enquiry : ***M.manipurensis, M.puriensis, M.clupeia***. Species to be reexamined – *C.leiognathi & C.portblairensis*

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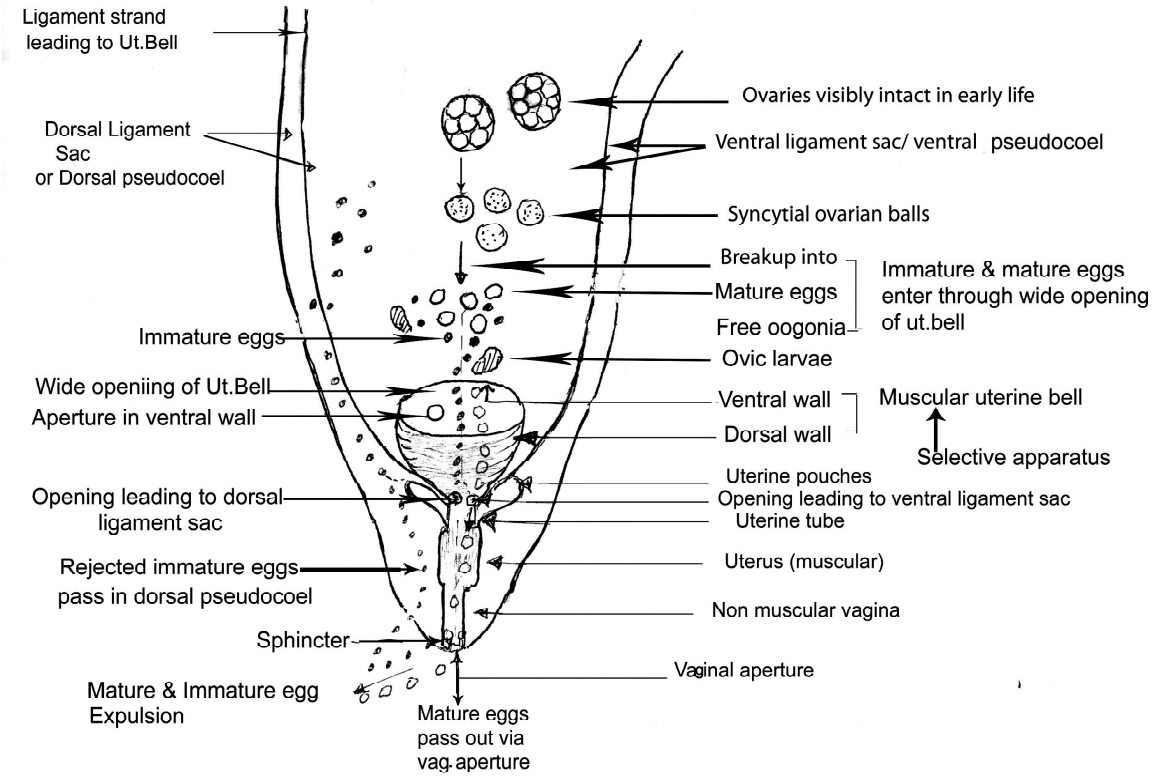
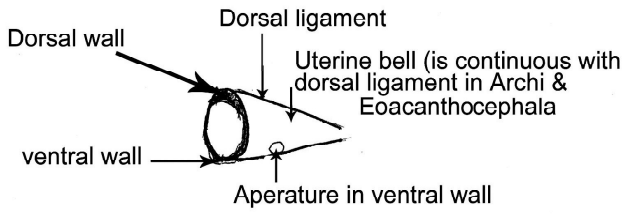
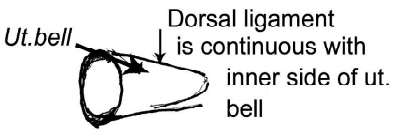


Fig.1.Reproductive organs of an acanthocephala (Diagrammatic)
Bhattacharya's openion of two openings at the post aspect of Ut.Bell. Yamaguti (1963)[3] objected about the immature egg elimination into Body cavity & mature eggs in the uterus.



Dorsal ligament association with uterine bell in *Archi & Eoacanthocephala*



Dorsal ligament association with uterine bell in *Palaeacanthocepha*

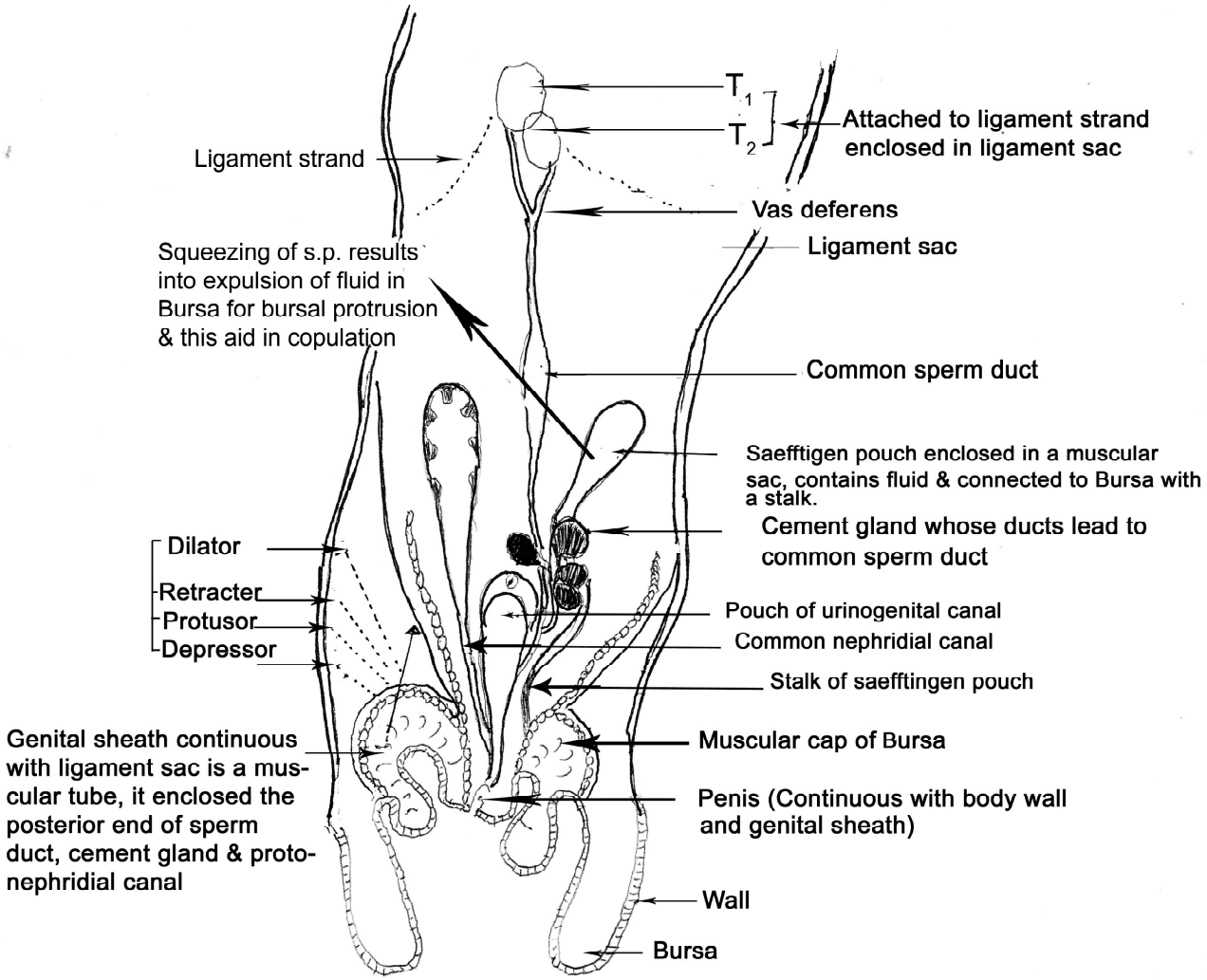


Fig.2. Male reproductive organs of an acanthocephala in Eoacanthocephala cement gland form a syncytial mass whose ducts lead to cement reservoir then its duct lead to sperm duct - secretion of cement gland seals the everted male Bursa around female gonopore

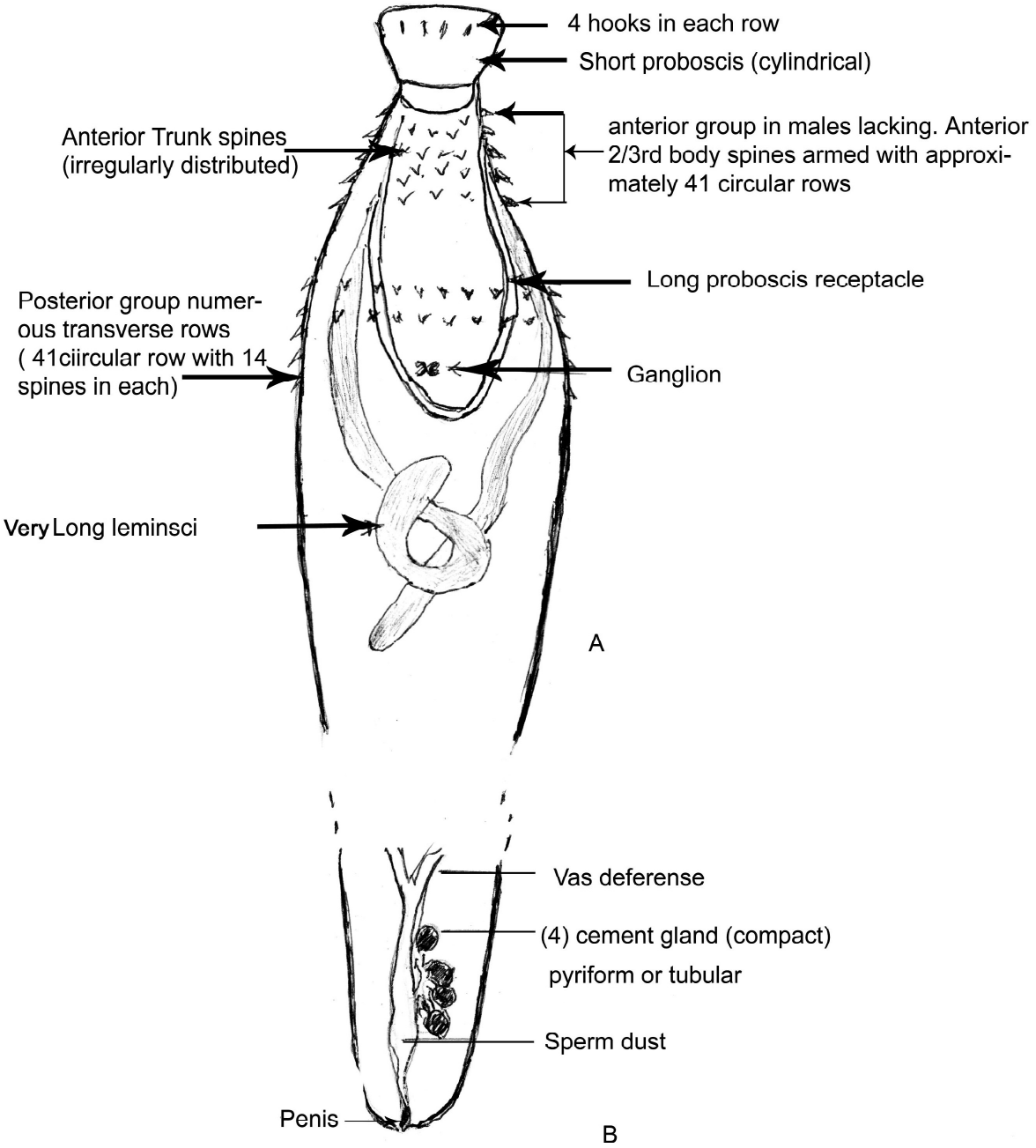


Fig.3. A. Diagrammatic sketch of Distribution of hooks on proboscis & trunk spines in *Cleaveius*

B. Cement gland (4) in number in *Cleaveius circumspinifer*

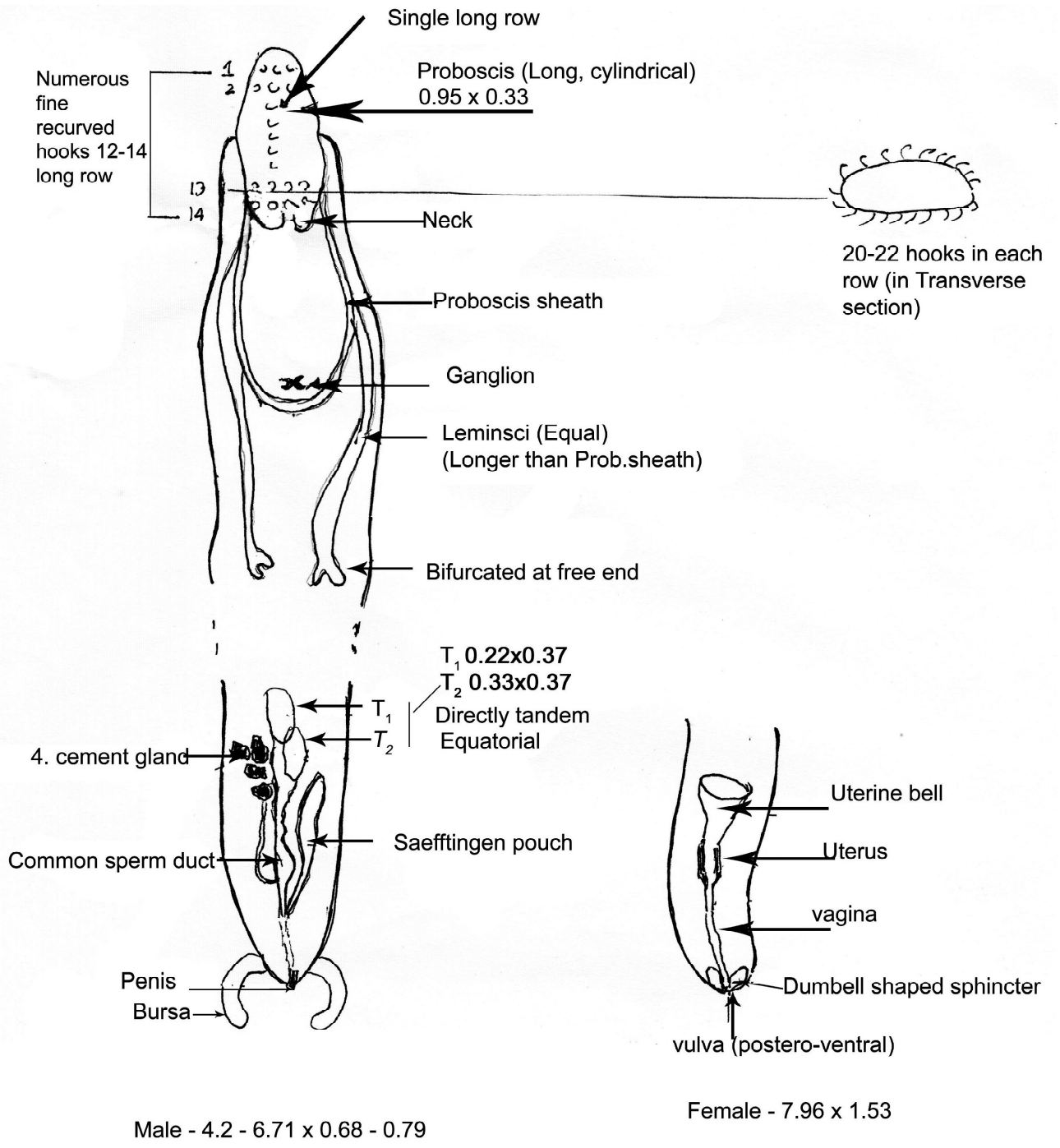


Fig.4. Diagrammatic sketch of M.*prashadi* Datta (1940) (male & post. part of female)